



Capstone Project Google Data Analytics

Cyclistic Bike-Share Analysis

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Agenda

1. **Company Overviews**
2. **Problem Definition & Planning**
3. **Data Cleaning Process**
4. **Data Visualization & Analysis**
5. **Recommendations**



A warm-toned photograph of a person with long hair writing in a spiral notebook on a wooden desk. A white mug sits on a saucer to the left, and a laptop is partially visible in the foreground. The text 'Company Overviews' is overlaid in white, with a green horizontal line below it.

Company Overviews

Cyclistic, Corp.

Company Overviews

Cyclistic, Corp.

Cyclistic is a bike-sharing company based in Chicago.



Launching 2016



Features over
5,824 bicycles

Promotes inclusivity
with reclining bikes,
hand tricycles, and
cargo bikes.



692 docking
stations

Bikers are flexible for
unlocking and
returning at any
stations in Chicago.



Pricing
Flexibility

Casual riders and
Cyclistic member.

Problem Definitions & Planning



Scope of Problem

- Cyclistic marketing strategy focus on broad consumer appeal with three pricing options:
 - 1) Single ride
 - 2) Full day pass
 - 3) Annual Membership
- Financial analysis indicates annual membership are more profitable

Customer Type

- Customer who purchase single ride or full-day pass are **Casual Bikers**.
 - Customer who purchase annual membership are **Cyclistic Member**.
-

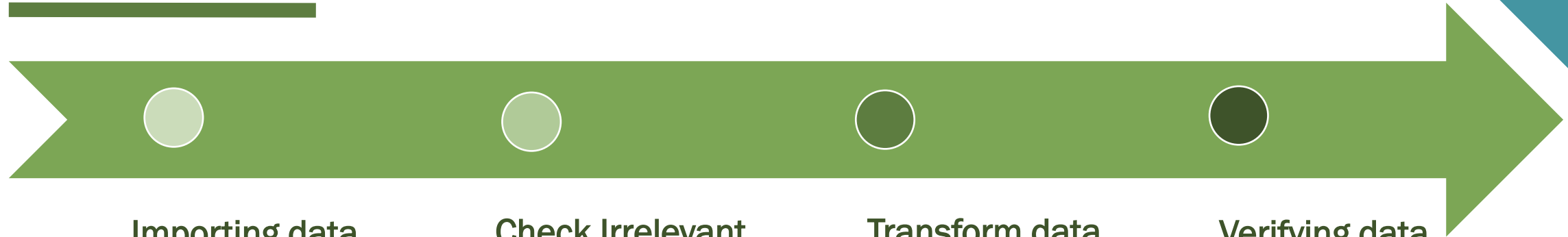
Future Growth Strategy

- Maximizing **Cyclistic Member** for future growth
- Convert **Casual Bikers** to **Cyclistic Member**

A close-up, sepia-toned photograph of a person's hand cleaning a desk surface with a white paper towel. The person is wearing a light-colored sweater with three buttons visible at the cuff. In the background, a computer keyboard and a blurred figure of another person are visible. The text "Data Cleaning Process" is overlaid in white, bold font, with a green horizontal line underneath it.

Data Cleaning Process

Data Cleaning Process



Importing data

- Collecting the data from January 2024 to December 2024
- Import the data from CSV file to an Excel spreadsheet

Check Irrelevant data

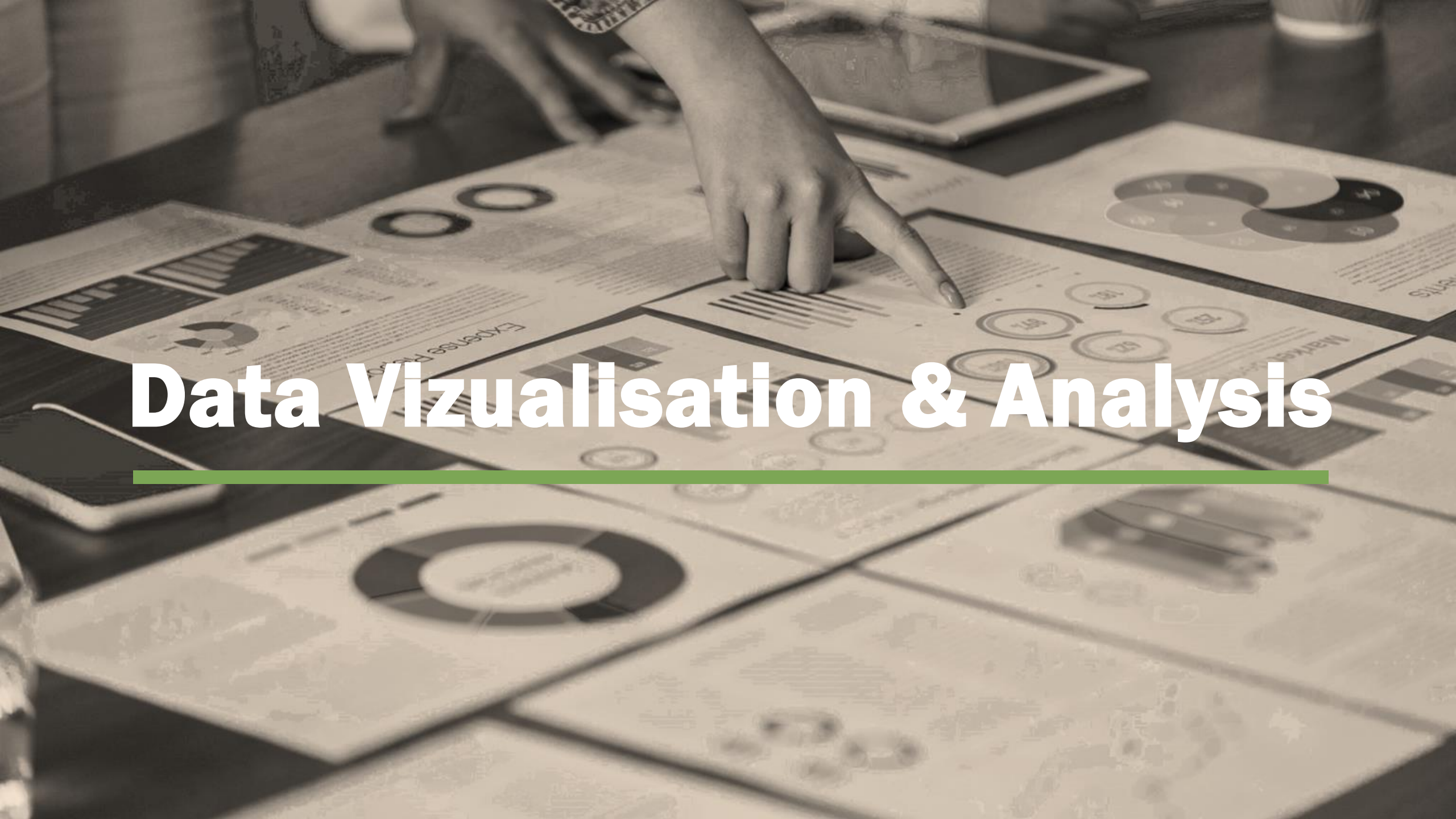
- Check irrelevant data in start date column
- Find whitespace in all column
- Find missing data in start station id column

Transform data

- Deleting irrelevant data
- Deleting extra whitespace
- Deleting column as not used in analysis

Verifying data

- Ensure all the column have correct data types
- Review a random sample of dataset

A hand is pointing at a large sheet of paper covered in various data visualizations. The paper features several charts, including bar charts, pie charts, and line graphs. The background is a blurred office setting with a tablet and a cup visible.

Data Visualisation & Analysis

Total Trips by Customer Type

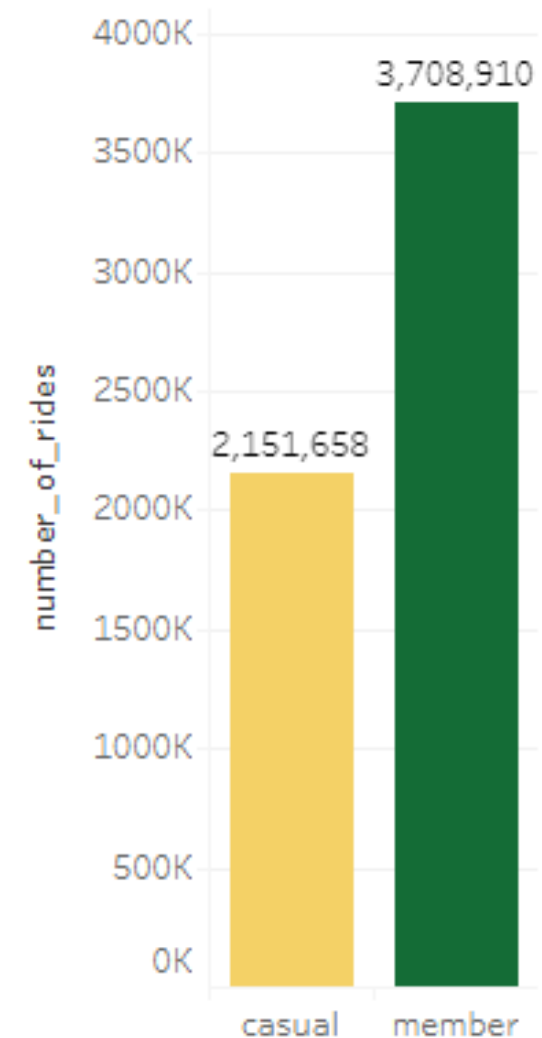
Insight

- Cyclistic members took the most rides with 63.3%
- Casual bikers accounted for the remaining 36.7%

Recommendation

- Highlight the benefits of membership, such as cost savings and unlimited ride options, for casual riders who frequently use the service.

Total Trips by Customer



Trip Duration by Customer Type

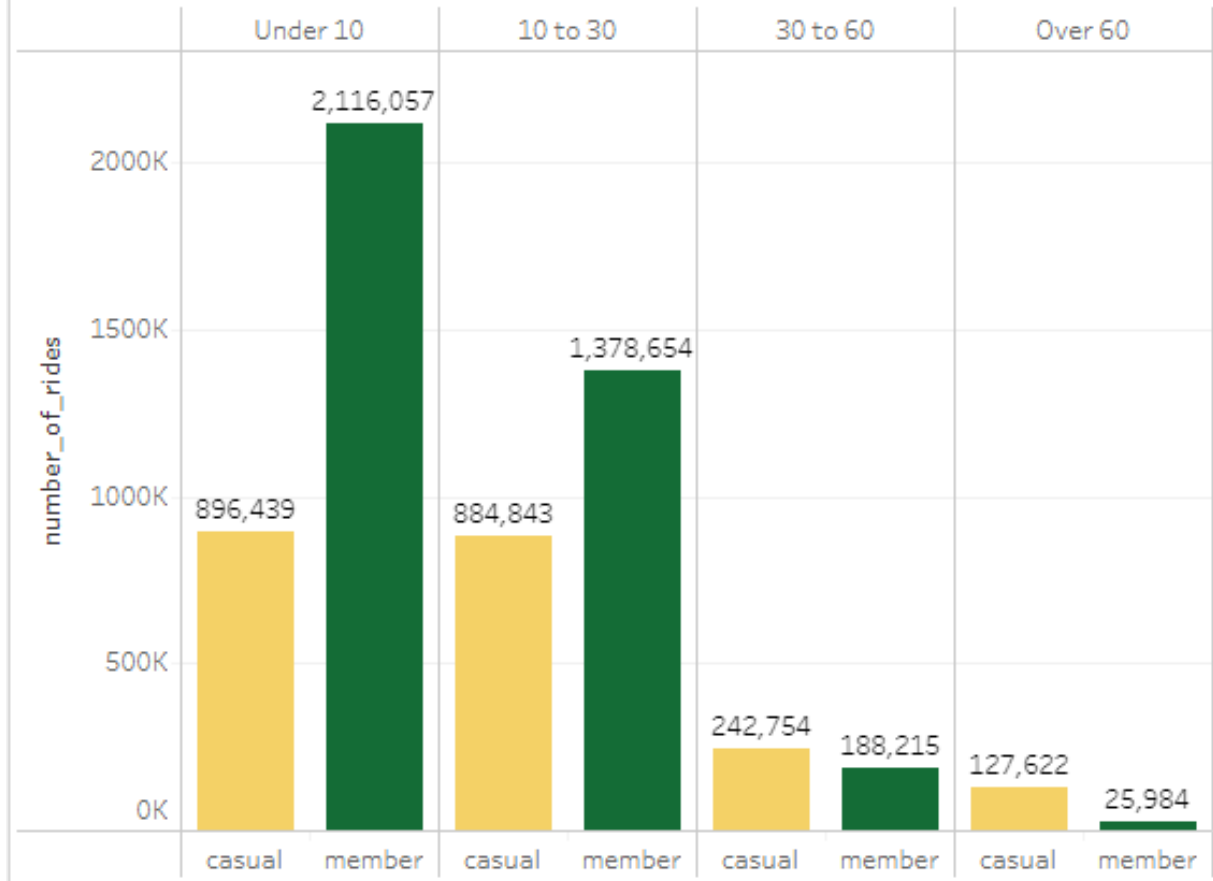
Insight

- Both of **cyclistic members** and **casual bikers** rides **under 10** minutes the most.
- **Casual bikers** take more rides than **cyclistic members** for trips **longer than 30** minutes.

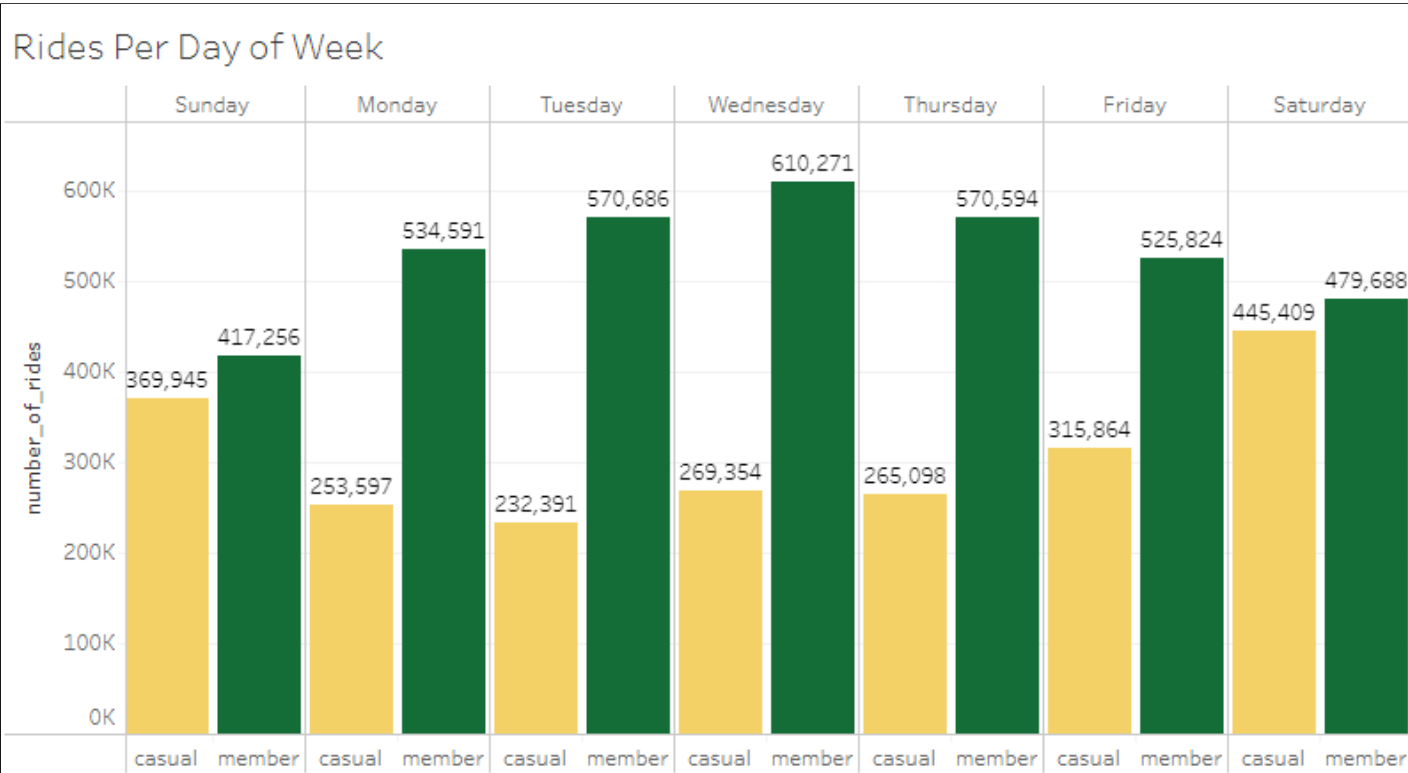
Recommendation

- **Develop benefits** of membership for **short rides**, such as reduced costs for frequent short trips, to appeal to casual users who prefer shorter durations
- **Offer membership incentives** for long-duration riders, such as capped daily or monthly charges, to show casual users the cost savings for trips over 30 minutes.

Trip Duration (in minutes)



Rides Per Day of Week



Insight

- **Cyclistic Members** have a higher number of rides on weekdays (**Monday to Friday**). This suggests that members primarily use the service for regular activities or commuting to work.
- **Casual riders** dominate on weekends (**Saturday and Sunday**). This indicates that casual users likely use the service for leisure or recreational purposes.

Recommendation

- **Introduce weekday membership** benefits for casual users, such as lower rates for rides during off-peak hours (e.g., before 10 a.m. or after 5 p.m.) when they upgrade to a cyclistic member.
- **Expand** service coverage or bike availability on **Wednesdays and Saturdays**, as they are the peak days for members and casual users.

Total Trips by Customer Type Per Month in 2024

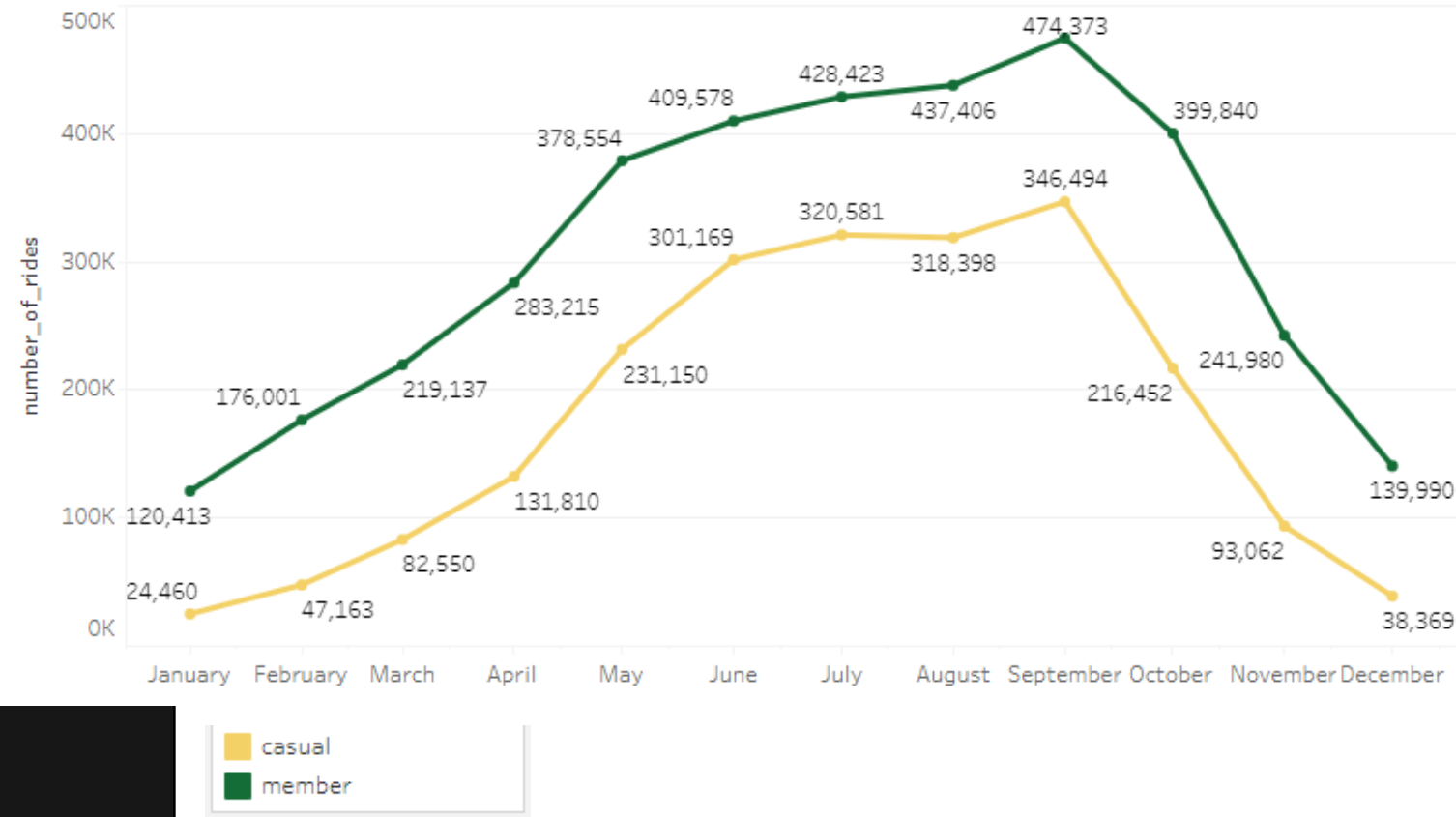
Insight

- Cyclistic members and casual bikers show **increased activity during warmer months** (likely due to better weather and outdoor activities).
- The **gap** between members and casual users is **smallest in summer months**, suggesting more recreational use from both groups

Recommendations

- **Focus marketing efforts** for casual users **during spring and summer** (e.g., vacation packages, discounts for group rides).
- **Develop strategies** to convert casual riders to members **during colder months** (e.g., offering indoor cycling partnerships, seasonal membership discounts, or exclusive winter benefits)

Rides per Month in 2024



Total Trips by Bike Type

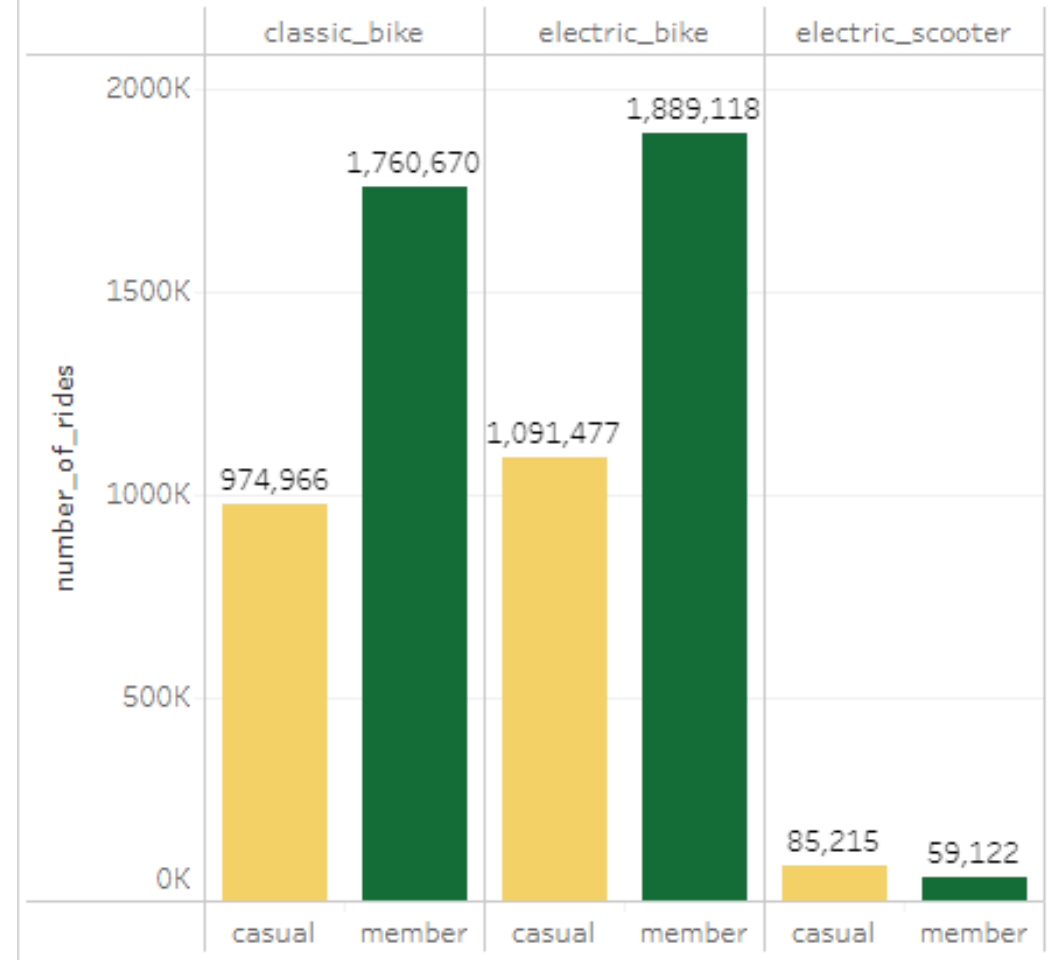
Insight

- **Electric bikes** are the **most popular** among both user groups, with **cyclistic members** having the highest count.
- **Casual bikers** use **electric scooters** more than **cyclistic members**.

Recommendation

- Offer subscription upgrades or bundles for **electric bike** access to casual users, encouraging conversions to membership
- **Create promotional campaigns** or incentives (e.g., discounts, loyalty points) to encourage members to use electric scooters.

Total Rides by Bike Type





Recommendation

Recommendation

Marketing Team

- **Develop strategies** to convert casual riders to cyclistic members **during colder months** (e.g., offering indoor cycling partnerships, seasonal membership discounts, or exclusive winter benefits).
- **Tailor marketing campaigns** to convert casual riders to cyclistic member **on weekends** (e.g., weekend packages or group rides).
- Run a **"Try Membership for a Week" program** to let casual users experience the benefits of being a member.

Customer Relationship Management

- **Introduce weekday membership** benefits for casual riders, such as lower rates for rides during off-peak hours (e.g., before 10 a.m. or after 5 p.m.), when they upgrade to a cyclistic member
- **Expand** service coverage or bike availability on **Wednesdays and Saturdays**, as they are the peak days for members and casual users.



Thank you

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